

Step 1: Calculate the Mean

$$\bar{x} = \frac{3.2 + 4.0 + 5.5 + 4.8}{4} = \frac{17.5}{4} = 4.375$$
$$\bar{y} = \frac{15 + 18 + 25 + 20}{4} = \frac{78}{4} = 19.5$$

Step 2: Calculate $(x_i - \bar{x})(y_i - \bar{y})$

x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	Product
3.2	15	-1.175	-4.5	5.2875
4.0	18	-0.375	-1.5	0.5625
5.5	25	1.125	5.5	6.1875
4.8	20	0.425	0.5	0.2125

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = 12.25$$

Step 3: Covariance

Using population covariance:

$$\text{Cov}(X, Y) = \frac{12.25}{4} = 3.0625$$

Answer:

- Mean of $x = 4.375$
- Mean of $y = 19.5$
- **Covariance = 3.0625**

2. Data Smoothing using Equal-Frequency Binning (5 Marks)

Given Data (27 values)

110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180

Divide into 3 Equal-Frequency Bins (9 values each)

Bin 1:

110, 112, 115, 115, 118, 120, 120, 122, 124

Mean:

$$\frac{110 + 112 + 115 + 115 + 118 + 120 + 120 + 122 + 124}{9} = \frac{1056}{9} = 117.33$$

Smoothed Bin 1:

117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33

Bin 2:

124, 126, 126, 126, 126, 130, 135, 135, 138

Mean:

$$\frac{1166}{9} = 129.56$$

Smoothed Bin 2:

129.56, 129.56, 129.56, 129.56, 129.56, 129.56, 129.56, 129.56, 129.56

Bin 3:

138, 138, 138, 140, 145, 150, 152, 160, 180

Mean:

$$\frac{1341}{9} = 149.00$$

Smoothed Bin 3:

149, 149, 149, 149, 149, 149, 149, 149, 149

Final Answer

- **Bin 1 Mean = 117.33**
- **Bin 2 Mean = 129.56**
- **Bin 3 Mean = 149.00**

These are the smoothed values using **equal-frequency (equal-depth) binning by bin means**.

3. Equal-Width Binning for Risk Categorization (5 Marks)

Given Data (18 BMI values)

18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5

Step 1: Compute Bin Width

- Minimum = **18.5**
- Maximum = **39.5**
- Range = **39.5 – 18.5 = 21**
- Number of bins = **3**

$$\text{Bin Width} = \frac{21}{3} = 7$$

Step 2: Equal-Width Bins

- **Bin 1:** 18.5 – 25.5
- **Bin 2:** 25.5 – 32.5
- **Bin 3:** 32.5 – 39.5

Step 3: Replace Values with Nearest Bin Boundary

Bin 1

18.5, 19.2, 21.8, 24.9, 25.5

→ **18.5, 18.5, 18.5, 25.5, 25.5**

Bin 2

26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4

→ **25.5, 25.5, 25.5, 32.5, 32.5, 32.5, 32.5, 32.5**

Bin 3

33.1, 33.6, 34.7, 38.2, 39.5

→ **32.5, 32.5, 32.5, 39.5, 39.5**

4. Data Reduction using Bin Medians (Medical Supply Costs) (5 Marks)

Given Data

120, 150, 160, 160, 140, 145, 145, 148, 150, 150, 155, 155, 155, 155, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300

Step 1: Sort Data 120, 140, 145, 145, 148, 150, 150, 150, 155, **155**, 155, 155, 160, 160, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300

Step 2: Divide into 3 Equal-Frequency Bins (9 values each)

Bin 1

120, 140, 145, 145, **148**, 150, 150, 150, 155

Median = **148**

Smoothed Bin:

148, 148, 148, 148, 148, 148, 148, 148, 148

Bin 2

155, 155, 155, 160, **160**, 165, 170, 170, 175

Median = **160**

Smoothed Bin:

160, 160, 160, 160, 160, 160, 160, 160, 160

Bin 3

175, 175, 175, 180, **190**, 200, 210, 230, 300

Median = **190**

Smoothed Bin:

190, 190, 190, 190, 190, 190, 190, 190, 190

5. Standardization of Patient Age Data (5 Marks)

Given Ages

18, 22, 25, 42, 28, 43, 33, 35, 56, 28

(a) Mean Absolute Deviation (MAD)

Mean:

$$\bar{x} = \frac{18 + 22 + 25 + 42 + 28 + 43 + 33 + 35 + 56 + 28}{10} = 33$$

Absolute deviations:

15, 11, 8, 9, 5, 10, 0, 2, 23, 5

$$\text{MAD} = \frac{15 + 11 + 8 + 9 + 5 + 10 + 0 + 2 + 23 + 5}{10} = \frac{88}{10} = 8.8$$

Answer: MAD = 8.8

(b) Z-score for the First Four Observations

Mean = **33**

Standard deviation (population):

$$\sigma = \sqrt{\frac{\sum(x - \bar{x})^2}{10}} \approx 10.84$$

Using

$$z = \frac{x - \bar{x}}{\sigma}$$

Age	Z-score
18	-1.38
22	-1.02
25	-0.74
42	0.83

Final Answers

- **Mean = 33**
- **MAD = 8.8**
- **Z-scores (first four ages):**
 - 18 → **-1.38**
 - 22 → **-1.02**
 - 25 → **-0.74**
 - 42 → **0.83**